

Hardware Sensor Test Documentation: Fingerprint Sensor

Module: Fingerprint Sensor Testing

Functionality: Enrollment and Registration

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_FP_001	Verify successful registration of a new fingerprint.	Secure lock screen password (PIN/Pattern) is configured. No prior fingerprints added.	1. Go to Settings → Password & Security → Fingerprint. 2. Follow prompts to place and lift index finger repeatedly. 3. Register edges of the finger.	The system must map the biometric templates incrementally and display a “Fingerprint added successfully” confirmation.	Pass
TC_FP_002	Verify behavior when attempting to enroll the same finger twice.	An index finger is already enrolled as “Fingerprint 1”.	1. Start a new fingerprint enrollment process. 2. Attempt to scan the same index finger.	The sensor subsystem must detect duplicate minutiae data and reject the process with a toast message: “This fingerprint is already enrolled”.	Pass
TC_FP_003	Verify maximum limit validation for fingerprint profiles.	System maximum enrollment threshold is reached (e.g., up to 5 fingers).	1. Attempt to select “Add Fingerprint” option when the limit is reached.	The “Add Fingerprint” button must be grayed out or prompt an alert indicating the maximum capacity has been met.	Pass

Functionality: Authentication and Security Bounds

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_FP_004	Verify fast unlocking with a registered fingerprint.	Device is locked. Ambient display or screen is turned ON.	1. Place the registered finger lightly on the physical/In-display sensor module.	The device must match the hardware matching algorithms and instantly unlock the device within nominal latency (<350ms).	Pass
TC_FP_005	Verify security rejection with an unregistered fingerprint.	Device is locked.	1. Place a non-enrolled finger on the sensor.	The screen must remain locked, trigger a mild haptic rejection buzz, and display an error message: "Fingerprint not recognized".	Pass
TC_FP_006	Verify brute-force lockout threshold execution.	Device is locked.	1. Intentionally authenticate with a wrong finger 5 consecutive times.	The system must temporarily lock out biometric authentication for 30 seconds and force the user to unlock via backup PIN/Pattern.	Pass

Functionality: Environmental Factors & Partial Inputs

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_FP_007	Verify sensor response to wet or sweaty fingers.	Fingerprint profile is enrolled normally.	1. Apply a minor layer of moisture/sweat to the finger surface. 2. Attempt to unlock the device.	Depending on sensor hardware tech (Optical vs Ultrasonic), it should gracefully retry or prompt "Ensure sensor surface is dry".	Pass
TC_FP_008	Verify partial scan matching capacity (Edge inputs).	Fingerprint edges were registered during initial setup.	1. Place only the extreme outer tip or edge of the finger on the sensor.	The system algorithm must resolve the partial input using available edge templates and unlock successfully.	Pass